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TOWARDS SUSTAINABLE ERP SYSTEMS: EMERGING TRENDS, CHALLENGES, AND FUTURE PATHWAYS

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Abstract:
Sustainable Enterprise Resource Planning (S-ERP) systems are essential for integrating sustainability into business operations and improving environmental, social, and economic performance for any organization. Nevertheless, the implementation of S-ERP systems is complicated and presents multiple challenges. The purpose of this review is to fill the knowledge gap by thoroughly analyzing the current status and utilization of S-ERP systems through an in-depth examination of existing literature to identify critical factors for successful implementation and offer guidance to support organizations in aligning sustainability goals with ERP systems. Employing a qualitative narrative review methodology, this research examines literature from major academic databases and focusing on studies published between 2000 and 2024. The research identifies key factors crucial to the successful integration of S-ERP systems, such as comprehensive strategic planning, efficient data management, strong managerial commitment, interdisciplinary expertise, and a carefully structured implementation plan. These factors not only streamline the initial deployment process but also ensure the system's flexibility to accommodate evolving sustainability goals and technological advancements. The study emphasizes the importance of a systematic approach to S-ERP adoption, with a focus on aligning with the organization's sustainability objectives and structural context to optimize benefits while minimizing risks. This research provides actionable guidance for organizations seeking to implement S-ERP systems effectively, offering strategies to navigate both the opportunities and challenges in this rapidly evolving field. It also delivers critical insights for researchers, industry practitioners, and organizational leaders, equipping them with the knowledge needed to enhance sustainability efforts through the adoption and optimization of S-ERP systems.

Key words: Enterprise Resource Planning, Sustainable ERP, Sustainability, Sustainable Business Processes, Sustainable Information Systems

INTRODUCTION

Enterprise Resource Planning (ERP) systems consist of software modules designed to automate and streamline corporate operations by enabling the real-time exchange of shared information, data, and processes across the entire organization. At the core of an ERP system is a central database that gathers and distributes data to multiple applications, supporting various functions within the company through a unified IT and information structure. This enhances the flow of information across the organization. ERP systems integrate different organizational functions, allowing data to be entered once and immediately updated across the business. This enables companies to manage resources more effectively by providing a comprehensive, integrated solution for their information-processing needs [1, 2]. Nevertheless, traditional ERP systems have primarily focused on enhancing financial and operational efficiencies, often overlooking broader sustainability considerations. While financial optimization has been a core objective of these systems, the complexities of sustainable development call for a more comprehensive approach. To support sustainability performance and regulatory compliance, ERP systems must provide transparency across financial, environmental, and social indicators [3].

Due to the rising focus on sustainable development, there has been an increase in interest regarding the creation and use of S-ERP systems. S-ERP systems are specifically developed to integrate sustainability activities across organization value chain, including environmental, social, and economic aspects [4, 5]. The implementation of S-ERP systems is complex and requires a master plan comprising a roadmap, framework, and guidelines [6, 7]. Hence, these systems seek to offer all-encompassing solutions that allow firms to efficiently monitor and report on their sustainability performance, therefore connecting their business operations with wider sustainability objectives [7, 8, 9].

S-ERP systems offer several benefits compared to conventional ERP systems. It boost organizational efficiency by seamlessly integrating processes and automating manual operations, hence optimizing productivity and resource management [10]. It facilitate initiatives that promote sustainability by offering tools for quantifying and documenting sustainability performance, including monitoring carbon footprints and optimizing resource utilization. These systems facilitate organizations in achieving enhanced resource management, adhering to sustainability standards, and making a positive contribution to environmental sustainability [3, 11]. Nevertheless, S-ERP requires ongoing support, maintenance, and updates to provide lasting benefits. However, conventional ERP systems may not prioritize as much importance on sustainability. Continuous development is crucial for aligning with evolving sustainability standards, incorporating advanced functionalities like carbon footprint and energy usage monitoring, and ensuring the successful achievement of advantages. In order to effectively manage the complex nature of S-ERP systems, it is essential to employ a methodical strategy that ensures a proper equilibrium between operational efficiency and sustainability goals. Regular updates and maintenance help companies integrate these systems into their overall sustainability strategy, enabling them to adapt to evolving challenges and opportunities. Maintaining this dedication is crucial for ensuring enduring value and effectiveness in advancing environmental objectives [12, 13].

However, there is a lack of study in the field of S-ERP systems, particularly concerning their deployment strategies and the key factors that contribute to their success. Most research has focused on traditional ERP systems, with only a few studies exploring the specific challenges and opportunities associated with integrating sustainability into ERP frameworks [3, 14, 15].The lack of research in this area highlights the necessity for a thorough examination that specifically focuses on the unique features of S-ERP systems and offers valuable insights into the strategies for their deployment. Therefore, this study aims to conduct a comprehensive analysis of existing literature to assess the current concepts, findings, and research fields related to the deployment of S-ERP systems. The study will analyze the factors that determine and limits of progress, as well as gaps in current studies. The research aims to address these gaps by offering a comprehensive understanding of how S-ERP systems can be effectively implemented and leveraged to attain long-lasting benefits in various organizational settings. The main goal is to ascertain the present state of the S-ERP system. The research question is: "What is the current state of S-ERP systems and what factors contribute to their success or pose challenges?" sustainability objectives

This study adopted a structured narrative review methodology to examine the deployment and utilization of S-ERP systems. Relevant literature was sourced from major academic databases, including Scopus, Web of Science, and IEEE Xplore. The search was conducted using specific keywords such as "Sustainable ERP," "S-ERP systems," and "Sustainability in ERP," targeting peer-reviewed articles published between 2000 and 2024. The selected timeframe reflects the evolution of ERP systems in response to increasing sustainability demands. This approach ensures the inclusion of diverse perspectives and comprehensive insights into the challenges, opportunities, and critical factors influencing the successful implementation of S-ERP systems. By synthesizing qualitative insights, this methodology aligns with the study's objectives of offering practical guidance and addressing the gaps in the current understanding of S-ERP systems.

The following sections provide a comprehensive exploration of S-ERP systems, detailing their historical development, components, benefits, and implementation challenges. The discussion also covers the strategic phases of S-ERP implementation, from planning through to post-implementation, highlighting the importance of each phase in achieving successful outcomes. Additionally, it emphasizes the factors that are pivotal for the effective implementation of S-ERP systems.

CONCEPTS OVERVIEW

Historical Development of ERP Systems

In the 1960s, the incorporation of computers began the emergence of companies creating software specifically designed for inventory management, material procurement, and production. Inventory control evolved into a systematic approach for overseeing business operations. In the 1970s, companies implemented Materials Requirements Planning (MRP) systems to manage purchasing, forecasting, and production scheduling. This development eventually gave rise to the establishment of organizations such as SAP and J.D. Edwards. J.D. Edwards updated the MRP system by adding closed-loop scheduling, increased shop floor reporting, and forward scheduling, which is referred to as MRP-II. These enhancements were aimed at reducing overhead expenses. In the late 1980s, technology became integrated into daily operational decision-making, leading to the emergence of prominent ERP providers like as SAP, IBM, J.D. Edwards, Baan, PeopleSoft, and Oracle. ERP software enhanced the clarity and accessibility of information on inventory and production levels, so enabling firms to achieve a competitive advantage.

During the 1990s, firms aimed to merge operational and accounting operations by implementing ERP systems in response to increasing competition. The term ERP, which originated by the Gartner Group, resulted in substantial expansion for six main vendors of business applications. The Y2K issue in 2000 led to increased adoption of ERP systems due to aggressive marketing efforts, resulting in a growth in the number of ERP vendors. The dotcom boom of 2001 resulted in instability, which in turn led to labor layoffs. In the late 2000s, Oracle purchased J.D. Edwards and PeopleSoft, while Infor Global Solutions acquired Baan and IBM's MAPICS. As a result, SAP, Oracle, and Infor emerged as the leading vendors in the ERP market.

Cloud computing has led to the advancement of ERP systems, as it allows for the movement of essential hardware off-site, resulting in reduced IT expenses. The 2016 ERP Report conducted by Panorama Consulting revealed a significant 40% increase in the use of cloud solutions compared to on-premise solutions. This shift can be given to a decrease in misunderstandings surrounding cloud computing. The growing emphasis on enhanced application security is driving an increasing number of firms to migrate to cloud-based ERP platforms. In addition, Lean concepts have been incorporated into ERP platforms to reduce operational inefficiencies. Scientists are currently working on the development of S-ERP systems for the purpose of managing sustainable operations [16, 17].

Current study offers important perspectives into the future trajectory and advancements of ERP systems, emphasizing a significant shift propelled by the increasing technology and developing company needs. Moreover, it is crucial to highlight the integration of cloud and edge computing into ERP systems, particularly in the context of the Industrial Internet of Things (IIoT) and intelligent factories. These results indicate a notable era of transformation and advancement for ERP systems [18]. Similarly, [19] investigates the utilization of ERP systems within the framework of Industry 4.0, with a specific emphasis on implementing cutting-edge technological solutions and generating prospects for further study. Furthermore, the research undertaken by [20] investigates the possible incorporation of blockchain technology into ERP systems. The text examines the current limitations and highlights potential areas of future study to enhance security and transparency. These research suggest that future ERP systems will adopt contemporary technology such as cloud computing, IIoT, and blockchain. This will result in the development of more effective, reliable, and adaptable solutions that are compatible with evolving corporate contexts.

The Concept of Sustainability in Information Systems

Organizations have a responsibility to adapt their business strategies due to government policies that enforce sustainability and the growing demand from customers for sustainable products. Companies must include sustainability efforts into their operations and actions in order to get sustainable results. Dependable data and information are necessary for the implementation of sustainability measures and the creation of precise sustainability reports. Nevertheless, the process of sustainability reporting and performance evaluation is frequently conducted manually through the use of spreadsheets. This approach is not efficient or effective in producing regular reports for stakeholders. With the increasing importance of sustainability in strategy and operations, a new wave of enterprise technologies is being developed to improve business processes. Integrating sustainability data, procedures, and reporting into S-ERP systems allows practitioners to effectively coordinate sustainable business operations. The S-ERP system optimizes the collecting, computation, assessment, and reporting procedures for sustainable business activities [21].

Conventional ERP systems have significantly enhanced and optimized company operations and management in several domains, including manufacturing, inventory management, distribution, purchasing, finance, human resources, customer relationship management, and sales. However, conventional ERP systems does not have the ability to evaluate sustainability indicators, such as the amount of resources used in the production process and its environmental impact [4]. Achieving sustainability in a firm requires a holistic and integrated approach that encompasses not just the product and its production processes, but also the whole supply chains, including the manufacturing systems throughout different product life cycles. To achieve this, it is necessary to create more advanced models, establish criteria for evaluating sustainable performance, and implement optimization methods at the product, process, and system levels [6, 22]. Current information systems lack in their ability to gather sustainability-related data and fail to offer sufficient support for managing data across all aspects of sustainability. A substantial proportion of this data may have qualitative information, which might pose difficulties in its handling and analysis. Hence, it is crucial to incorporate sustainable business processes and procedures. During the procurement process, firms need to establish connections between their data and information and stakeholders in order to efficiently monitor the sourcing of sustainable raw materials. Failure to develop integrated information systems hinders organizations from adopting a proactive strategy to ensure ongoing and early cooperation during the beginning phases of business needs research. Furthermore, the firms do not possess the capability to include sustainability into the standards for new items. As a result, there is frequently a disconnect between a company's economic performance and its management of environmental and social factors. The economic advantages of applying environmental and social management strategies are not well comprehended. Lack of integration can impede simultaneous improvements in the economic, environmental, and social aspects [6, 9].

To overcome this limitation, many manufacturers like as SAP, Oracle, and Microsoft are actively involved in developing S-ERP software solutions specifically designed to tackle this problem [9]. S-ERP systems are information systems that are driven by sustainability considerations and cover all stages of the value chain. S-ERP systems may be seen as holistic, integrated, and all-encompassing solutions for addressing sustainability-related business challenges [9]. This is a novel form of enterprise systems that aims to facilitate the integration of sustainable business units inside an organization. This information system facilitates firms in providing relevant and reliable data and information on the impact of sustainability on business [4].

S-ERP systems are frequently employed to manage and record company activities pertaining to environmental and social impact, as well as to align organizational processes, staff, and products with corporate sustainability goals and standards. The S-ERP system consists of two main scenarios: data collection and reporting. The system may gather data on sustainability key performance indicators for reporting, using both manual and automatic approaches. In order to streamline the reporting process, the required data may be transmitted using various interfaces, such as text, diagrams, tables, and charts, that allow comprehensive evaluation. Therefore, the technology would aid the senior executives in making prompt and long-term decisions. S-ERP systems have a beneficial influence on an organization's sustainability in several ways. The system primarily utilizes cloud-enabled technology, which in turn enables the efficient utilization of server capacity by freeing up location constraints. Moreover, it possesses the capacity to save energy and power because to its constant need to sustain a low temperature. Moreover, the utilization of S-ERP systems plays a crucial role in reducing the carbon footprint. Moreover, the system facilitates intra-organizational information transmission, hence reducing the need for excessive printing and minimizing waste. Moreover, the use of an S-ERP system may elevate stakeholder engagement by providing precise sustainability reporting and facilitating efficient collaboration and communication through information exchange [23]. A comparison of conventional ERP systems and S-ERP systems, highlighting their key differences, is provided in the comparison Table 1.

Table 1
Comparison of conventional ERP systems and S-ERP systems

Aspect	Conventional ERP Systems	(S-ERP) Systems
Primary Focus	Enhances operational efficiency in areas like manufacturing, finance, HR, inventory, and CRM.	Focuses on integrating sustainability into all business operations, including environmental and social impact management.
Sustainability Capabilities	Lacks the ability to track or report sustainability metrics like resource usage and environmental impact.	Optimizes the collection, evaluation, and reporting of sustainability metrics, such as energy consumption and carbon footprint.
Data Management	Primarily financial and operational data, often handled using manual processes such as spreadsheets for reporting.	Automates sustainability data collection and integrates data sources for comprehensive sustainability performance evaluation.
Sustainability Reporting	Relies on manual processes (e.g., spreadsheets), making reporting inefficient and prone to errors.	Uses automated and integrated reporting tools that offer real-time data through various formats (text, tables, charts, etc.).
Support for Environmental Impact	Minimal support for tracking environmental metrics.	Includes modules for tracking environmental impact, reducing carbon footprint, and monitoring energy usage.
Support for Social Impact	Limited to basic HR functions like employee turnover and diversity.	Capable of managing broader social sustainability factors, including community impact, employee well-being, and diversity.
Stakeholder Engagement	Limited support for stakeholder engagement and sustainability-related decision-making.	Enhances stakeholder engagement with accurate sustainability reporting and facilitates collaboration through shared data.
Technological Integration	Not designed to handle new sustainability data sources or integrate with sustainability-focused IT systems.	Integrates sustainability data, processes, and reporting across the entire value chain using cloud technology.
Energy Efficiency	Does not focus on energy consumption or environmental impact.	Cloud-enabled technology helps reduce energy use, optimize server capacity, and minimize waste (e.g., reducing paper use).
Decision Support	Primarily supports operational and financial decision-making.	Helps senior executives make informed sustainability-related decisions by providing comprehensive performance evaluations.
Barriers	Lack of support for environmental and social metrics, inefficient reporting processes, and limited integration capabilities.	Challenges include incorporating qualitative data, aligning sustainability with business processes, and resource limitations.
Examples of Providers	SAP, Oracle, Microsoft (with conventional ERP solutions).	SAP, Oracle, Microsoft (developing specific S-ERP solutions targeting sustainability).

Hence, by incorporating environmental, social, and economic factors into conventional ERP frameworks, S-ERP systems offer a transformative approach to managing sustainability within businesses. S-ERP systems, compared to traditional ERP systems, provide the automated gathering, assessment, and reporting of sustainability data, including energy usage and carbon footprint. These solutions, which make use of cutting-edge technology, help businesses better engage stakeholders, expedite sustainability reporting, and match corporate operations with sustainability objectives. Adoption of S-ERP systems becomes crucial in achieving a comprehensive, sustainable business plan as firms place a greater emphasis on sustainability.

KEY DIMENSIONS OF SUSTAINABLE ERP

S-ERP facilitate the incorporation of sustainable business operations, processes, and data into a unified platform, fostering a comprehensive approach to sustainability [8]. The S-ERP framework combines sustainability and decision-making paradigms to offer a holistic approach to system deployment. The sustainability paradigm has three dimensions: society, environment, and economy, ensuring that the S-ERP system incorporates every aspect of sustainability. The societal side emphasizes the importance of achieving social fairness and making a positive influence on the community. The environmental aspect aims to enhance resource efficiency and minimize ecological footprints. The economic aspect seeks to improve financial performance and promote sustainable economic growth.

The S-ERP framework encompasses not just the sustainability paradigm but also a decisional paradigm that addresses decision-making at strategic, tactical, and operational levels. Top management at the strategic level develops long-term goals and plans that are in line with the organization's sustainability objectives. The tactical level encompasses the responsibilities of middle management in formulating and executing plans that connect the strategic objectives with the daily operational activities. The operational level is primarily concerned with the day-to-day operation and choices required to carry out the tactical plans and accomplish the strategic goals. The S-ERP framework incorporates these paradigms to guarantee that sustainability is seamlessly incorporated into all levels of decision-making and business operations, ranging from high-level formulation of strategies to daily operational duties. By adopting a comprehensive approach, the company may successfully implement sustainable practices, resulting in enhanced sustainability performance and alignment with wider environmental and societal objectives [21].

Figure 1 illustrates the general dimensions of the S-ERP framework, depicting how the sustainability paradigm (society, environment, and economy) and the decisional paradigm (strategic, tactical, operational) are integrated within the system to ensure comprehensive sustainability performance and alignment with broader environmental and societal objectives.

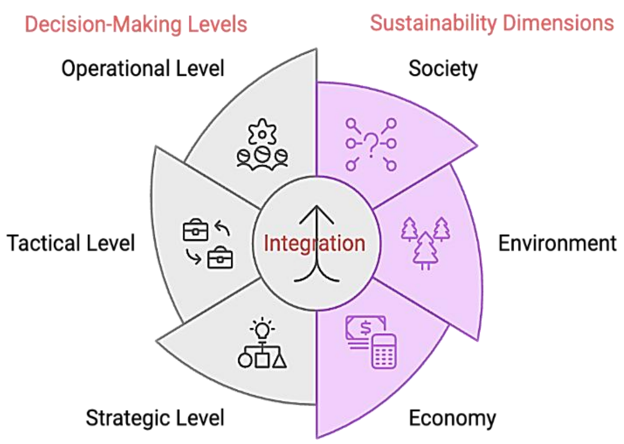

Fig. 1 S-ERP System Dimensions
Source: developed by authors.

Implementing S-ERP systems necessitates a well-planned strategy to manage the complexities and guarantee the seamless integration of sustainable practices inside a firm. The implementation road map for S-ERP systems is crucial for several reasons. Firstly, it offers a comprehensive and organized blueprint, defining every step and stage of the implementation process, hence minimizing uncertainty and aiding companies in quickly navigating complex processes. The roadmap guarantees the successful achievement of the set targets by linking the implementation with the organization's sustainability and business objectives. Additionally, it facilitates resource management by enabling effective scheduling and distribution of time, finances, and workforce, therefore guaranteeing that the project remains within the allocated budget and adheres to specified timelines. In addition, a roadmap serves to identify possible risks and obstacles at an early stage, allowing for the creation of strategies and plans to minimize the chances of project delays or failures. It improves stakeholder communication by presenting a distinct visual depiction of the strategy, guaranteeing that all stakeholders comprehend their roles and obligations, hence facilitating their endorsement. The plan further implements monitoring and control methods to oversee development and evaluate performance, guaranteeing that the project stays on track. This process guarantees adherence to applicable norms and standards by clearly outlining roles, duties, and deadlines, so ensuring accountability. Moreover, employing a methodical technique enables ongoing enhancement, enabling firms to gain insights from each stage of execution and refine procedures for subsequent endeavors. In summary, a roadmap is essential for effectively managing the intricacies of implementing an S-ERP system, which in turn leads to the successful incorporation of sustainable practices throughout the value chain [22].

According to [22], The S-ERP roadmap serves as a detailed guide for implementing S-ERP systems within organizations. The project is divided into three distinct phases: pre-implementation, implementation, and post-implementation. Each phase comprises several stages designed to ensure thorough and effective project development as shown in Figure 2.

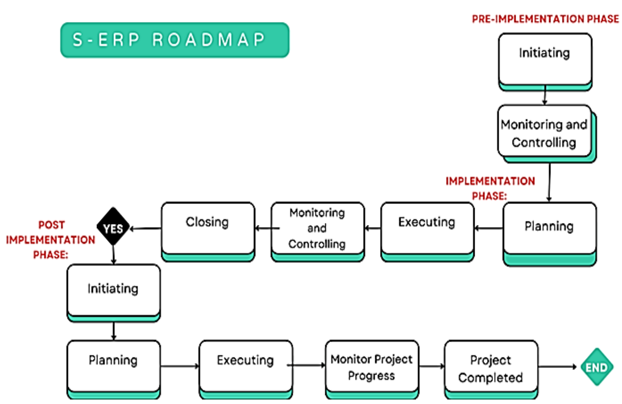

Fig. 2 S-ERP roadmap
Source: developed by authors based on [22].

- **Pre-implementation Phase**:
  - **Initiating**: This step includes obtaining the necessary authorization to commence the S-ERP implementation project. The purpose is to synchronize stakeholder expectations with the project goals by providing clarity on the business case and scope.
  - **Monitoring and Controlling**: responsible for overseeing and managing all preparation operations to ensure they are in line with the project's objectives.
- **Implementation Phase**:
  - **Planning**: In this stage, the project's general scope is identified, goals are defined and refined, and strategies are devised to fulfill the project objectives. The main objective is to determine and specify the trajectory, methods, and plan for the successful execution of the S-ERP implementation project.
  - **Executing**: the main objective is to carry out the tasks and activities that were planned to comply with the project's requirements.
  - **Monitoring and Controlling**: Continuous monitoring and controlling processes are carried out to track the progress and performance of the project, ensuring that it stays on course.
  - **Closing**: This stage involves the ongoing practice of tracking the development and performance of the project to ensure that it remains on track.
- **Post-implementation Phase**:
  - **Initiating**: This stage, similar to the pre-implementation phase, involves defining actions and plans for the system's go-live.
  - **Planning**: This phase entails creating a comprehensive strategy for the S-ERP system to be implemented, guaranteeing a seamless transition.
  - **Executing**: The activities that were planned in the preceding stage are implemented in order to accomplish the targeted goals.
  - **Monitoring and Controlling**: involve ongoing processes that oversee and regulate post-implementation operations to guarantee continual progress and alignment with company goals.
  - **Closing**: Concluding all actions after implementation to verify the system is completely functional and satisfies the users' requirements.

The roadmap integrates project management phases and process groups to ensure the continuous and effective deployment of S-ERP systems, emphasizing the need for a systematic and comprehensive approach to guide practitioners through each stage of the project.

CURRENT TRENDS IN SUSTAINABLE ERP

ERP systems are being more and more combined with other technologies or methods to improve business sustainability, especially in the discrete manufacturing sector as shown in Figure 3.

These systems are integrated with approaches such as lean management and lean green practices, which prioritize improved efficiency by reducing waste, maximizing resource use, and improving processes.

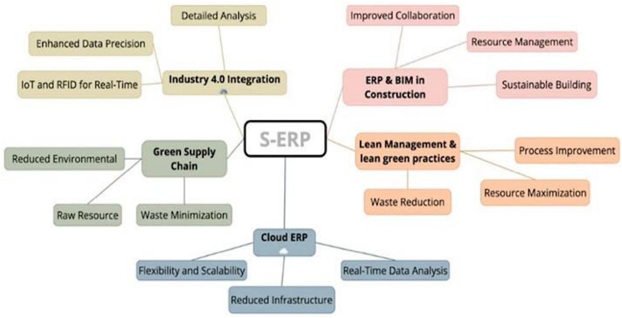

Fig. 3 ERP Systems Integration with Technology for Sustainability

This combination facilitates the coordination of production and distribution operations with demand, resulting in waste reduction and resource conservation. Moreover, Green Supply Chain Management (GSCM) aims to diminish waste, optimize the utilization of raw resources, and mitigate environmental consequences by implementing inventive procedures. Integrating GSCM with ERP systems enhances interdepartmental communication and collaboration with supply chain partners, resulting in enhanced coordination and more efficient resource allocation. Cloud technology and Big Data applications enhance the efficiency and sustainability of ERP systems by offering flexibility, scalability, and real-time data analysis. In addition, Industry 4.0 technologies, such as the Internet of Things (IoT) and Radio-Frequency Identification (RFID), are combined with ERP systems to collect industrial data in real-time. This integration enhances data precision and facilitates detailed analysis and prediction. This integration provides extensive solutions that tackle the economic, environmental, and social aspects of sustainability. It presents significant prospects for improving corporate sustainability in the discrete manufacturing sector, where efficient resource management and waste reduction are vital for sustainable operations [24].

On the other hand, the use of Cloud ERP systems is connected to sustainability in several significant ways. From an environmental perspective, Cloud ERP solutions decrease the requirement for physical infrastructure, such as servers and data centers that are owned by the firm. As a result, this reduces carbon footprints and total energy usage. Cloud suppliers are able to achieve this decrease by employing energy-efficient technology and perhaps utilizing renewable energy sources. Cloud ERP systems facilitate efficient data management, allowing firms to improve their supply chains, minimize waste, and simplify procedures, therefore reducing their environmental footprint. From an economic perspective, Cloud ERP solutions provide cost effectiveness, enabling small and medium-sized firms (SMEs) to utilize them and promoting sustainable economic growth by directing resources towards innovation. Cloud ERP systems provide the capacity to scale up corporate operations without the need for substantial investments in IT infrastructure. This allows organizations to easily adjust to market fluctuations without incurring excessive expenses. Cloud ERP solutions promote social engagement among stakeholders, fostering sustainable business practices such as ethical sourcing and equitable labor standards. In addition, optimizing procedures and increasing productivity can boost employee contentment and welfare, so fostering a more favorable and enduring work atmosphere. Ensuring system and data security is essential for establishing and maintaining trust and dependability, which are fundamental for sustaining company operations. Adapting Cloud ERP systems to meet individual company requirements is crucial for aligning with sustainability objectives, facilitating optimal resource utilization, and advocating for sustainable practices. Hence, the effective integration of Cloud ERP systems may greatly aid an organization's sustainability initiatives by diminishing environmental harm, fostering economic development, and improving social sustainability [25]

Also, integrating ERP with Building Information Modeling (BIM) systems is a popular application that offers a revolutionary method to improving sustainability in building projects. This holistic solution transcends the individual benefits of deploying each system alone. The integration of ERP and BIM yields a substantial improvement in sustainability compared to their individual implementation. This integration utilizes the advantages of both technologies to enhance sustainability at every stage of construction projects. BIM plays a vital role in the design process by facilitating sustainable building designs through intricate performance assessments that enhance energy efficiency, decrease water consumption, limit carbon emissions, and use eco-friendly materials. Nevertheless, BIM alone is unable to uphold these sustainable standards during both the building and operating stages. ERP systems are crucial in integrating and optimizing company operations, resulting in cost reduction and quality improvement. As a result, they contribute to economic advancements, social fairness, and environmental preservation. The connection guarantees a smooth and uninterrupted transfer of information throughout all stages of the project, with BIM offering comprehensive design and material data that ERP use to effectively oversee resources during construction and maintenance. This comprehensive strategy minimizes waste, maximizes resource use, and guarantees the achievement of sustainability objectives across the whole process, from initial design to ongoing operation. In addition, the integration promotes enhanced collaboration among stakeholders, leading to improved decision-making and project efficiency, therefore enhancing social sustainability. ERP solutions enhance economic sustainability by reducing operating expenses and enhancing productivity, advantages that are further enhanced when integrated with BIM. Although there are difficulties in aligning the various architectures and functionality of the systems, customisation guarantees that processes are synchronized to gain substantial sustainability advantages. In addition, the integrated system facilitates ongoing monitoring and reporting of sustainability measures, guaranteeing that objectives are established, accomplished, and sustained in the long run. Hence, the integration of ERP and BIM offers a comprehensive, efficient, and sustainable solution for building projects that surpasses the use of both systems separately [11, 26].

In conclusion, the present developments in S-ERP systems encompass the integration of ERP with cutting-edge technologies like lean management, Green Supply Chain Management (GSCM), cloud technology, IoT, and RFID. This integration aims to boost the efficiency of resources, minimize waste, and promote operational effectiveness. This integration facilitates enhanced coordination of production and distribution with demand, resulting in substantial preservation of resources. Similarly, in the construction industry, the integration of S-ERP with BMI improves sustainability at every stage of a project by improving design, building, and operating processes. These integrations provide significant sustainability advantages, such as decreased environmental impact through lower carbon footprints and energy consumption, economic expansion by enabling scalable and cost-efficient operations, and enhanced social practices by promoting ethical sourcing and fair labor standards. Nevertheless, they also pose issues such as the requirement for ongoing adjustment to advancing technology, guaranteeing the security of systems and data, and aligning the systems with the unique sustainability goals of the company. Therefore, the effective execution of these integrated ERP systems may greatly enhance an organization's sustainability efforts, leading to enduring environmental, economic, and social advantages.

BENEFITS OF SUSTAINABLE ERP SYSTEMS

As shown in Figure 4 Adopting S-ERP solutions offers several advantages that improve the sustainability and effectiveness of enterprises.

These systems facilitate the incorporation of sustainable business operations, processes, and data onto a unified platform, fostering a comprehensive approach to sustainability. S-ERP systems enhance the precision and uniformity of sustainability data by consolidating data analysis and report creation, which is essential for making informed decisions and reporting to stakeholders. They adhere to the Triple Bottom Line (TBL) paradigm, which incorporates economic, social, and environmental aspects, ensuring a harmonious integration of sustainable practices into the fundamental company strategy. This alignment results in economic advancement, social fairness, and environmental preservation. In addition, S-ERP systems optimize corporate processes, leading to substantial cost savings and enhancements in quality, so promoting more sustainable and efficient operations. In addition, they offer assistance in adhering to regulatory mandates and sustainability criteria, including strong reporting features to ensure transparency and responsibility. Clear and accessible sustainability data facilitates improved communication and collaboration with stakeholders, fostering trust and supporting long-term objectives. Furthermore, S-ERP systems promote innovation by fostering the creation of sustainable goods and services, generating fresh market prospects, and improving competitive edge.

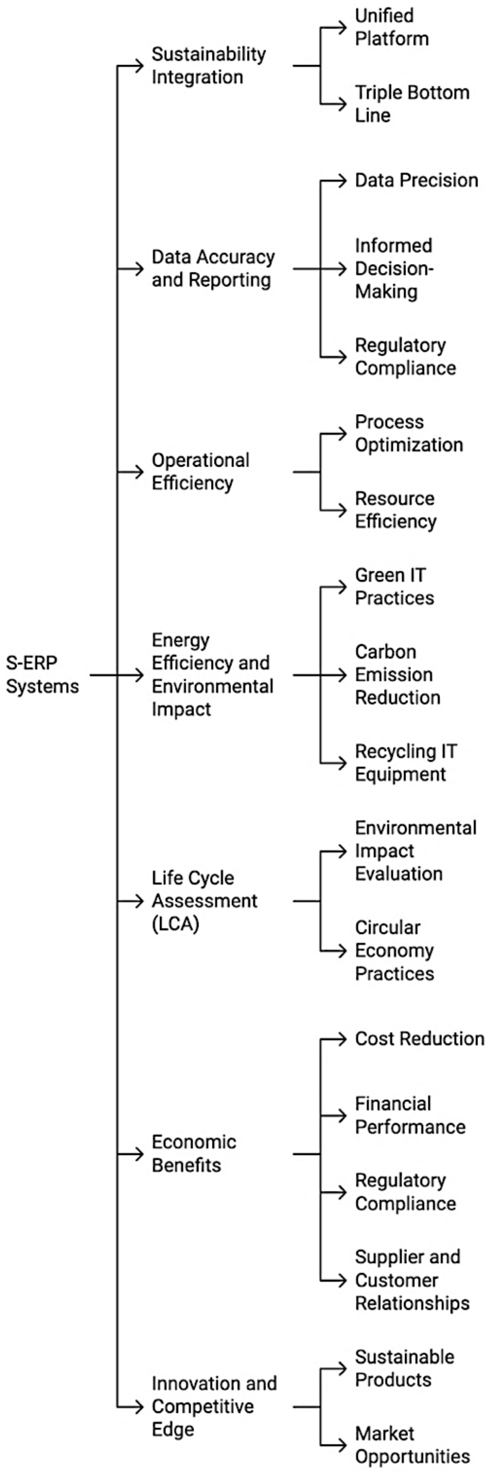

Fig. 4 Advantages of adopting S-ERP systems
Source: developed by authors.

Therefore, S-ERP systems provide a thorough method for overseeing sustainability, resulting in substantial advantages that enhance the efficiency, adherence to regulations, and competitive standing of a firm [8].

The use of S-ERP systems has a substantial influence on the long-term viability of a company. S-ERP systems facilitate the integration of sustainable processes, information, and data throughout all levels of the value chain in enterprises. This integration facilitates the efficient management of sustainable business practices within enterprises, thereby resolving the problem of segregation that often occurs between suppliers, business departments, and consumers during the implementation of sustainability initiatives. Prominent IT corporations like as Oracle, Microsoft, and SAP have created S-ERP solutions that adapt to these requirements. Implementing S-ERP effectively helps optimize sustainable company operations, lower expenses, enhance quality, and eventually foster economic advancement, social fairness, and environmental preservation, therefore aligning with the TBL of sustainability [27].

S-ERP systems offer a multitude of environmental advantages by using green IT practices to promote responsible corporate operations. These systems enhance energy efficiency by optimizing energy usage and using strategies such as server virtualization and automated power management. In addition, they promote for the utilization of sustainable resources, minimize the generation of waste and pollution, and foster the practice of recycling and repurposing IT equipment. S-ERP solutions enable enterprises to efficiently manage their sustainability performance by consolidating data and activities across business functions, therefore reducing the environmental impact of their operations. Moreover, S-ERP solutions allow organizations to adhere to environmental requirements, diminish their carbon emissions, and endorse sustainable development projects in diverse industries [28]. Moreover, the use of environmentally friendly IT practices in S-ERP systems, such as server virtualization and automated power control, enhances energy efficiency and diminishes carbon emissions. Therefore, S-ERP systems have a significant impact on promoting sustainable business operations by integrating environmentally friendly practices into the organization's fundamental processes and IT infrastructure [29].

Also, the use of S-ERP systems makes a substantial contribution to environmental sustainability by incorporating Life Cycle Assessment (LCA) into company operations. This integration enables firms to methodically evaluate and control the ecological effects of their goods and activities at every stage of their lifespan. The main environmental advantages include increased resource efficiency by optimizing resource utilization and minimizing waste, decreased environmental impact by monitoring and reducing emissions and energy consumption in real-time, and enhanced decision-making through comprehensive environmental performance data. In addition, S-ERP systems assist organizations in adhering to environmental rules, therefore minimizing the likelihood of non-compliance and the resulting fines. They advance sustainable practices by integrating environmental factors into fundamental business procedures, fostering the creation of environmentally friendly goods and green technology. In addition, S-ERP systems facilitate circular economy activities by monitoring the whole lifespan of items, encouraging recycling, reuse, and waste reduction. This helps to preserve natural resources and minimize environmental deterioration. Incorporating LCA into S-ERP systems offers a strong structure for firms to efficiently handle their environmental footprints, resulting in enhanced sustainability performance and contributing to worldwide environmental objectives [30].

From another perspective, the implementation of an S-ERP system provides several economic advantages to enterprises. S-ERP systems greatly improve transparency and efficiency in administrative activities, such as finance management, warehousing, transportation, and sales, by integrating different processes, organizational structures, marketing strategies, and operational platforms. By unifying, the overall cost of operations and administrative overhead is reduced. For instance, in a case study, the introduction of a consolidated ERP system led to a 20% decrease in overall expenses and a substantial improvement in time effectiveness, lowering procedures by one-fifth compared to the previous condition [29]. A study showed a direct correlation between the adoption of S-ERP systems and economic sustainability performance, suggesting the possibility of cost-saving advantages. In addition, S-ERP systems promote financial performance by delivering up-to-date and precise information that improves decision-making abilities, resulting in more effective resource management and higher returns on assets (ROA), equity (ROE), and profitability. By incorporating contemporary management methodologies like lean manufacturing and total quality management (TQM) into S-ERP systems, there is an additional enhancement in cost reduction and efficiency optimization. Furthermore, S-ERP systems facilitate adherence to regulatory mandates, mitigating the possibility of monetary sanctions and bolstering the organization's standing. Furthermore, they enhance connections with suppliers and consumers by enhancing transparency and communication along the supply chain, resulting in a potential rise in market share and competitive advantage. Therefore, the use of S-ERP systems can result in significant economic benefits, such as decreased operating expenses, superior financial outcomes, improved decision-making capabilities, and a more advantageous market position [31].

In conclusion, S-ERP solutions provide substantial advantages through the integration of sustainable business operations, processes, and data into an integrated platform. These systems improve the accuracy of data, facilitate well-informed decision-making, and adhere to the TBL framework, which promotes economic, social, and environmental sustainability. Their objective is to enhance efficiency, minimize expenses, and enhance quality, all while guaranteeing adherence to regulatory requirements. Nevertheless, S-ERP systems pose difficulties such as the requirement for ongoing adjustment, data protection, and substantial initial capital expenditure. Despite these difficulties, S-ERP systems have the potential to generate significant environmental, economic, and social advantages when they are adopted and maintained efficiently.

CHALLENGES AND BARRIERS

As shown in Figure 5, Current ERP systems have deficiencies in many crucial areas when it comes to fulfilling the requirements for sustainability-related capabilities.

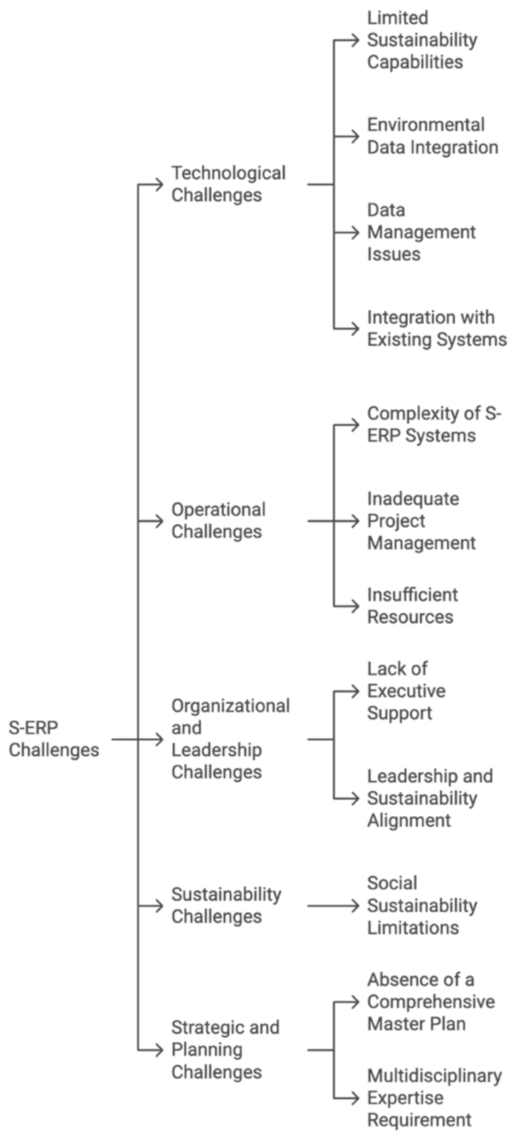

Fig. 5 Challenges of adopting S-ERP systems
Source: developed by authors.

Existing ERP systems are mainly developed to enhance financial and operational effectiveness, with an emphasis on reducing costs, financial planning, and managing resources. However, they are not intrinsically capable of adequately addressing broader sustainability measures. For instance, conventional accounting techniques such as First-In, First-Out (FIFO) are included to maximize financial results, but they may not take into consideration environmental factors such as waste minimization. While certain ERP systems have started to include environmental tracking capabilities like monitoring energy consumption, analyzing carbon footprint, and managing waste, these features are often restricted and not fully integrated into essential functions.

As a result, businesses need to look for supplementary tools or customized solutions to obtain comprehensive environmental sustainability metrics. Current ERP systems have limited development in the social elements of sustainability. Although HR modules may monitor some social indicators like as employee turnover and diversity, they do not possess the necessary depth to effectively handle wider social sustainability projects. For example, it is not widely supported to measure certain social performance criteria, such as the influence on the community, the well-being of employees, and diversity beyond basic demographic information [15].

The use of S-ERP systems may be noticed in ERP systems that are complex and require significant effort. Numerous firms have difficulties while installing ERP systems, and a few have even had failures. The failures can be linked to causes such as poor support from senior executives, insufficient allocation of funds, incompetent project management, user-related concerns, and inconsistencies and errors in data. The complexity of S-ERP systems has grown as a result of incorporating new types of data, new sources of data, and including more stakeholders. establishing S-ERP is anticipated to be more challenging than establishing a regular ERP system, as it necessitates addressing three sustainability factors: social, environmental, and economic. The utilization of S-ERP systems has the potential to equal or surpass the performance of prior ERP systems. This would significantly influence the attainment of sustainability goals and objectives [22].

Moreover, a significant obstacle is the absence of a thorough and all-encompassing master plan that incorporates a roadmap, structure, and rules. In the absence of this guidance, businesses frequently face elevated implementation expenses, prolonged schedules, and an augmented likelihood of failure. In addition, the management of sustainability data presents considerable obstacles owing to inconsistencies, errors, and the complexity of combining many data sources into a cohesive platform. Conventional data management techniques, like as spreadsheets, are insufficient for handling the intricacies of S-ERP systems. Another significant obstacle is the requirement for robust support and dedication from senior leadership, along with enough resources. Numerous implementation endeavors encounter difficulties as a result of inadequate administrative backing and limited financial resources. Moreover, the absence of efficient project management is a significant issue, since inadequate planning and risk management techniques frequently result in project setbacks and failures. Implementing S-ERP involves a combination of technical, management, and sustainability understanding, which can be tough for firms that lack the requisite skills and competence. Integrating S-ERP systems with current business processes and IT infrastructure also poses substantial challenges. When sustainability programs and information systems plans are not properly aligned, it can lead to problems with integration and inefficiency. The presence of these hurdles highlights the need of having a well-organized master plan, comprehensive methods for managing data, strong backing from leadership, efficient practices for project management, and knowledge from several disciplines in order to guarantee the successful installation of an S-ERP system [8].

In order to overcome the challenges associated with deploying S-ERP systems, companies have the option to embrace many potential solutions and best practices. The integration of green IT practices with ERP systems improves environmental sustainability by enhancing energy efficiency through the implementation of server virtualization, storage consolidation, automated power management, and the utilization of thin clients. Organizations have to establish green procurement strategies that give priority to energy-efficient and environmentally-friendly IT equipment, and choose vendors with robust environmental practices. It is essential to promote the recycling and reuse of IT equipment. By implementing programs for this objective, it can extend the life cycles of equipment and decrease waste

Furthermore, obtaining strong support from leaders is essential in pushing for projects that promote sustainability and allocating the required resources. Utilizing resilient project management approaches such as Agile assists in effectively handling the complicated nature of S-ERP systems. Utilizing sophisticated data management solutions guarantees the incorporation of various data sources and upholds data integrity. It is essential to form multidisciplinary teams consisting of experts in technology, sustainability, and business in order to ensure the successful implementation of S-ERP. Finally, by organizing frequent workshops, the IT and sustainability teams may coordinate their efforts and collaborate towards shared sustainability goals, therefore promoting a culture of environmental responsibility inside the organization [28].

Additionally, the development of an S-ERP master plan concept offers a comprehensive method to address issues that may arise during S-ERP implementation initiatives. This comprehensive strategy tackles the inherent complexities of incorporating sustainability into conventional ERP systems. The master plan guarantees the thorough incorporation of sustainability practices into ERP systems by providing a well-organized framework. This framework encompasses all stages of the project, including initiation, execution, and monitoring. The package consists of a comprehensive plan that clearly defines the essential actions and procedures, a structure that connects sustainability goals with corporate activities, and instructions that offer practical measures for professionals to follow. This comprehensive strategy not only simplifies the implementation process but also minimizes risks, expenses, and time, thereby improving the probability of attaining sustainability objectives and assuring the long-term success of the S-ERP system. The master plan's strategic alignment with project management principles enhances the efficient use of resources, interaction with stakeholders, and continual development. This makes it an essential tool for enterprises seeking to incorporate sustainability into their core operations [8].

Lastly, In order to successfully apply an S-ERP framework, businesses must guarantee comprehensive integration by including sustainability into all levels of decision-making, ranging from strategic to operational. This may be achieved by utilizing sustainability indicators to assess performance across environmental, social, and economic aspects. Ensure that the framework is in accordance with the organization's strategic objectives and consistently revise these objectives to incorporate changing sustainability criteria. Utilize resilient project management approaches, such as phase-gate procedures and project management systems, to ensure the project stays on schedule. Establish an interdisciplinary team of individuals with proficiency in sustainability, information technology, and business management. Ensure that they get ongoing training to stay informed of the most recent developments in their respective fields. Implement ongoing feedback channels, regularly perform audits and evaluations to improve and strengthen the S-ERP framework, guaranteeing its continued effectiveness and relevance [21].

To conclude, the implementation of S-ERP systems is a difficult and resource-intensive process that encounters several problems, including data discrepancies, lack of executive support, insufficient funding, inadequate project management, and data inaccuracies. The inclusion of novel data types, sources, and stakeholders adds further complexity to the process. An significant obstacle is the lack of a comprehensive master plan, resulting in elevated expenses, extended timelines, and increased chances of failure. Traditional methods of managing data are inadequate, and there is often a lack of strong backing from leadership. In order to address these difficulties, firms should use green IT practices to enhance energy efficiency, develop green procurement strategies, and encourage the recycling of IT equipment. Effective leadership and robust project management, such as Agile, are essential. Successful implementation requires the use of advanced data management technologies and the collaboration of interdisciplinary teams. Facilitating regular workshop to align IT and sustainability initiatives promotes teamwork in working towards common objectives. Creating an S-ERP master plan provides a thorough structure for incorporating sustainable practices, minimizing risks, expenses, and time, and guaranteeing the long-term effectiveness of the system. Continuous feedback mechanisms and frequent audits serve to enhance the framework's efficacy and pertinence.

TECHNOLOGICAL INTEGRATION WITH S-ERP

As organizations prioritize sustainability, the integration of advanced technologies has become essential to the development and functionality of S-ERP systems. Unlike conventional ERP systems, which primarily focus on operational efficiency, S-ERP systems incorporate environmental, social, and economic factors to align with sustainability goals [32, 33]. Advanced technologies such as cloud computing, the IIoT, artificial intelligence (AI), and blockchain enable S-ERP systems to effectively manage sustainability metrics, streamline data collection and analysis, and provide transparent, real-time insights. This technological integration allows S-ERP systems to meet the complex demands of modern sustainability initiatives, making them invaluable tools for organizations seeking to minimize their environmental footprint while maintaining economic performance [34, 35].

Cloud computing plays a crucial role in enabling S-ERP systems by providing scalable data storage, processing, and accessibility across multiple locations [33]. It supports real-time data collection and analysis, reducing costs and enhancing flexibility[36]. Cloud-based ERP systems offer resource elasticity, ease of use, and improved sustainability performance [37]. They facilitate collaborative data sharing for sustainability reporting and resource management [38]. Cloud computing contributes to environmental sustainability by improving energy efficiency and reducing carbon emissions [39]. The integration of cloud computing with Business Intelligence (BI) and ERP applications offers optimal solutions for efficient business operations [40]. Cloud computing technology supports sustainable development by balancing environmental, economic, and social benefits [41].

On other hand, IIoT enables S-ERP systems to connect with physical devices, sensors, and machinery, gathering real-time data on energy usage, emissions, and other sustainability metrics [41, 42]. IIoT facilitates real-time monitoring of production processes, optimizing resource usage and minimizing environmental impact [43]. It enables preventive maintenance, improves operational efficiency, and supports data-driven decision-making [44, 45]. The implementation of IIoT in manufacturing processes leads to increased sustainability, reduced costs, and improved overall performance [43]. However, challenges such as data privacy and security must be addressed for successful implementation [46].

Also, AI and machine learning (ML) are revolutionizing sustainable development and ERP systems by enhancing data analysis, pattern identification, and predictive capabilities [47, 48]. These technologies enable predictive maintenance, reducing waste and improving efficiency across industries [49, 50]. In supply chain management, ML applications optimize supplier selection, risk prediction, and demand forecasting, contributing to sustainability and circular economy initiatives [51]. Big data analytics, combined with ML, help reduce carbon footprints by optimizing transportation routes, enhancing energy efficiency, and improving inventory management [52]. AI-driven solutions in smart production promote sustainable manufacturing practices [53]. These advancements in data-driven sustainability offer powerful tools for addressing environmental challenges and promoting ecological resilience across various sectors [54].

Blockchain technology offers significant potential for enhancing sustainability in supply chain management and reporting. It ensures data transparency, traceability, and security, which are essential for trustworthy sustainability reporting [55, 56]. Blockchain can track sustainable sourcing and ensure accountability across the supply chain, aligning with responsible business practices [57, 58]. The technology addresses challenges in implementing sustainable practices by creating transparency, real-time information sharing, and data security [59]. Integration of blockchain with ERP systems can further optimize supply chain functions and build trust in a trustless environment [60] For sustainability reporting and assurance, blockchain enhances trust, transparency, and traceability, meeting stakeholder demands and potentially leading to uniform standards in evaluating sustainability reports

[61]. Despite its promise, blockchain adoption faces barriers that require further research and development [56]. The combined benefits of these technologies help organizations align their operations with sustainability goals, meet regulatory requirements, and provide stakeholders with accurate, real-time sustainability reports. Furthermore, these technologies make S-ERP systems more agile, enabling organizations to quickly adapt to new sustainability standards, market expectations, and environmental challenges. As a result, organizations using S-ERP systems with integrated technology are better equipped to make informed, sustainability-focused decisions that benefit both the environment and their business performance.

CONCLUSION, FUTURE DIRECTIONS AND RESEARCH OPPORTUNITIES

The historical development and theoretical foundations of ERP systems highlight a remarkable evolution driven by technological advancements and business requirements. From early inventory management solutions in the 1960s to the sophisticated ERP systems of today, the journey reflects significant strides in integrating operational and accounting functions. The transition from MRP to MRP-II and subsequently to comprehensive ERP systems underscores the industry's efforts to enhance visibility, reduce costs, and gain competitive advantages. However, the continuous technological evolution, particularly the shift towards cloud computing and the integration of Lean techniques, indicates that ERP systems are far from reaching a plateau. The incorporation of modern technologies like cloud computing, IIoT, and blockchain promises to usher in a new era of ERP systems characterized by efficiency, security, and flexibility.

Despite these advancements, current ERP systems fall short in addressing sustainability comprehensively. Traditional ERP systems are primarily designed for operational and financial efficiency, lacking inherent capabilities to measure and manage sustainability metrics effectively. The emergence of S-ERP systems attempts to bridge this gap by integrating sustainability considerations into all stages of the value chain. S-ERP systems are poised to provide comprehensive solutions that address environmental, social, and economic dimensions of sustainability. They offer potential benefits such as improved resource efficiency, compliance with regulatory requirements, and enhanced stakeholder engagement. However, the implementation of S-ERP systems is fraught with challenges, including data management complexities, lack of strong managerial support, and the need for multidisciplinary expertise.

S-ERP systems are crucial for embedding sustainability into organizational functions, driving environmental, social, and economic performance. However, the implementation of S-ERP systems presents significant challenges, as noted in contemporary research. This paper provides a comprehensive review of the current status and critical factors for successful S-ERP adoption. The findings emphasize that strategic planning, robust data management, strong managerial support, multidisciplinary expertise, and a well-structured roadmap are essential for effective S-ERP implementation. These elements not only support the initial deployment but also ensure the system's adaptability to evolving sustainability goals and technological advancements. The review underscores the need for a systematic approach to S-ERP implementation, highlighting the importance of alignment with organizational sustainability objectives and structural requirements to optimize the benefits of S-ERP systems while mitigating associated risks.

Future opportunities in S-ERP systems are driven by advancements in several key technologies. Cloud and edge computing solutions offer enhanced scalability and flexibility in managing data and resources, reducing IT costs by offloading hardware requirements to the cloud, and improving data security and disaster recovery capabilities. The IIoT enables real-time monitoring and optimization of manufacturing processes, enhances predictive maintenance capabilities to reduce downtime, and improves supply chain management through real-time data integration. Blockchain technology increases transparency and traceability in supply chains, enhances data security and integrity, and streamlines compliance with regulatory requirements through immutable records. AI and machine learning provide advanced data analytics for better decision-making, automate routine tasks to free up resources for strategic activities, and offer predictive analytics to foresee and mitigate risks. Sustainability integration allows for comprehensive tracking and reporting of sustainability metrics, optimizes resource use to minimize environmental impact, and improves stakeholder engagement through transparent sustainability reporting. Enhanced user experience features include intuitive interfaces and dashboards for better user interaction, customizable workflows to meet specific business needs, and improved mobile accessibility for on-the-go management. Improved collaboration and communication tools offer enhanced internal and external stakeholder collaboration, real-time information sharing and decision-making, and integration with social media and other communication platforms. By embracing these emerging technologies and practices, future S-ERP systems can offer more efficient, secure, and sustainable solutions, aligning with the evolving needs of businesses and their commitment to sustainability.

While this study employs a qualitative narrative review methodology to synthesize insights from existing literature, focusing on practical guidance and highlighting key factors for successful S-ERP implementation, future research could benefit from incorporating bibliometric and systematic literature review methodologies. The narrative review approach allowed for the exploration and contextualization of diverse perspectives, providing a broad and flexible understanding of the challenges and opportunities in S-ERP systems. However, a bibliometric analysis could complement these findings by identifying key trends, influential authors, and foundational works that shape the field, adding a quantitative dimension. Similarly, a systematic literature review would bring additional rigor and structure to the analysis, offering a more exhaustive and quantitative understanding to further enhance the insights gained from this qualitative study.

In conclusion, while ERP systems have undergone significant transformations over the decades, the journey towards fully sustainable and integrated solutions is ongoing. The theoretical foundations and historical developments provide a solid basis for understanding current trends and future directions. However, realizing the full potential of S-ERP systems will require overcoming substantial challenges through strategic planning, robust data management, and multidisciplinary collaboration. The integration of modern technologies and sustainability paradigms into ERP systems promises a future where businesses can achieve operational excellence while fulfilling their sustainability commitments.

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